

Object Oriented Programming

BSIT - Fall 2024

Semester 3

Quiz - 2

Time Allowed: 30 Mins

Total Marks: 30

1. Multiple Choice Questions:

(8 marks)

1. What is the correct way to declare a function in C++ that takes no arguments and returns an integer?
 - a) `int func;`
 - b) `int func();`
 - c) `void func();`
 - d) `func int();`
2. Which of the following statements is true about recursion vs loops?
 - a) Every recursive function can be converted to a loop
 - b) Every loop can be converted to recursion
 - c) Recursion always has better performance
 - d) Both A and B
3. What is the output of the following function:

```
int f(int n) {  
    if (n <= 1) return 1;  
    return f(n - 1) + f(n - 1);  
}
```

 - a) Factorial
 - b) Fibonacci
 - c) Power of 2 generator
 - d) Sum of digits
4. Which of these is NOT possible in recursion?
 - a) Infinite recursion
 - b) Mutual recursion
 - c) Returning multiple values
 - d) Printing values
5. What will be the output:

```
#include <iostream>  
using namespace std;
```

```
int fun(int n) {
    if(n == 0) return 0;
    return n + fun(n - 2);
}
int main() {    cout << fun(7);    }
```

- a) 16
 - b) 12
 - c) 7
 - d) Infinite recursion
6. Which statement is correct about recursion?
- a) Recursion always requires more memory than loops
 - b) Every recursion must have a base case
 - c) Recursion cannot be replaced by iteration
 - d) Recursion always runs faster than loops
7. Which of these loops is guaranteed to terminate?
- a) while(1){}
 - b) for(;;)
 - c) do{ }while(true);
 - d) None of these
8. If a function in C++ has a parameter with a default argument, when is the default argument value assigned?
- a) At function definition time
 - b) At function declaration time
 - c) At function call time, when no argument is provided
 - d) At program compilation time

2. Write the output of the following code snippets:

(8 marks)

Code snippet	Output
<pre>int main () { for (int i = 1; i <= 3; i++) { for (int j = 1; j <= 2; j++) { cout << i << j << " "; } } return 0; }</pre>	

<pre>#include <iostream> using namespace std; int fun(int n) { if (n == 0) return 0; if (n % 2 == 0) return fun(n - 1) + n; return fun(n - 1) - n; } int main() { cout << fun(6); return 0; }</pre>	
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3. Rescue Bots

(4 marks)

A team of rescue bots reports their daily energy usage in units. Write a function that takes an array of energy values for **n** days and returns:

- the day with the highest energy usage, and
- the average usage.

You must not calculate both in the same function.

Q4. Test Your Logic**(3 marks)**

Write a **function** to compute the remainder when dividing two integers without using %.

Q5. Population Collapse**(7 marks)**

A strange species reproduces such that:

- Each year, every creature produces 2 new creatures.
- However, every third year, half the population dies.

Write a recursive function `population(years)` that returns the population after **n** years, starting with 1 creature.
